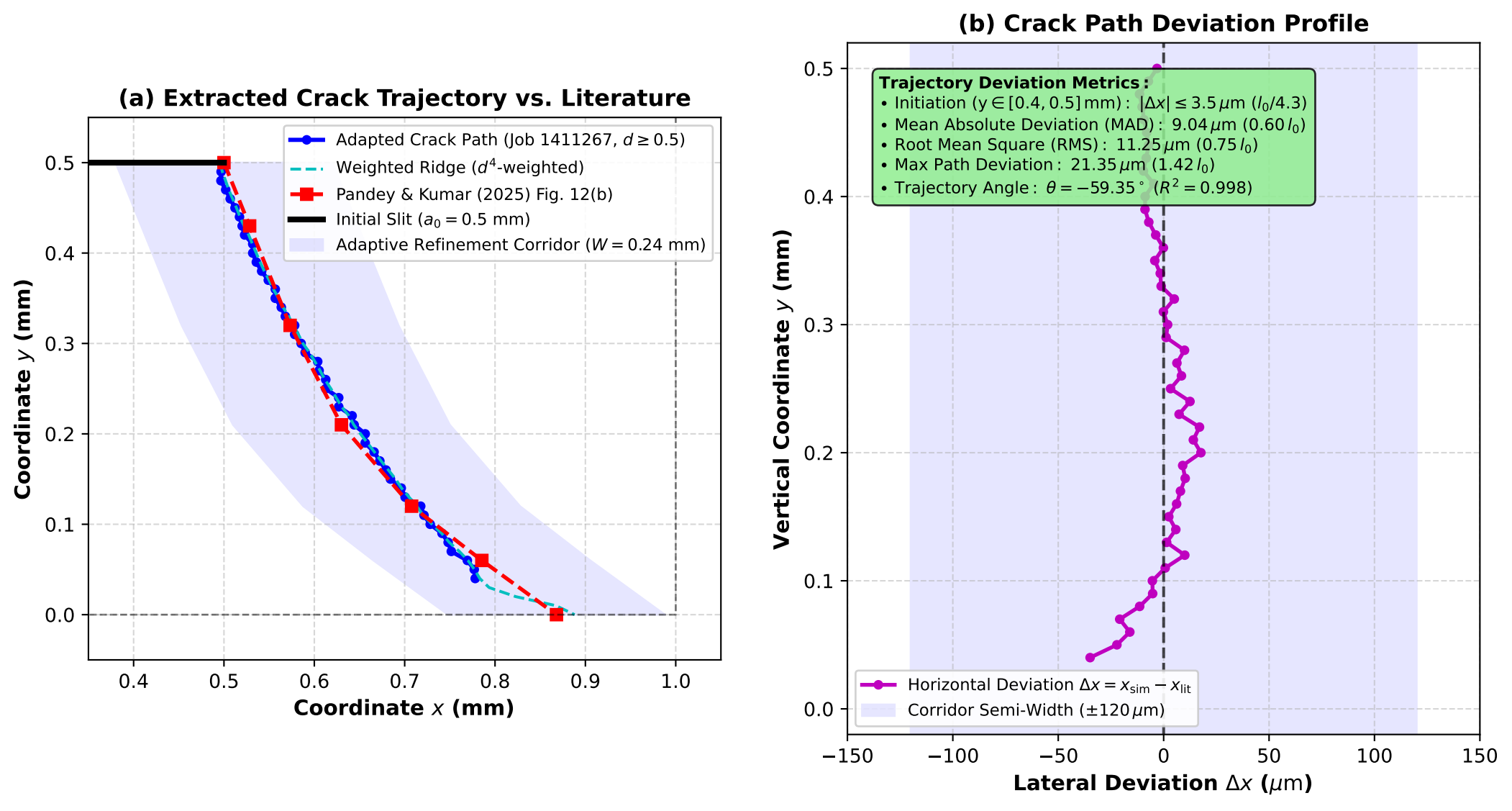
(a) Extracted Crack Trajectory vs. Literature
Adapted Crack Path (Job 1411267, $d \geq 0.5$)
Weighted Ridge (d^4-weighted)
Pandey & Kumar (2025) Fig. 12(b)
Initial Slit ($a_0 = 0.5$ mm)
Adaptive Refinement Corridor ($W = 0.24$ mm)
Coordinate y (mm)
Coordinate x (mm)
0.0
0.1
0.2
0.3
0.4
0.5
0.4
0.5
0.6
0.7
0.8
0.9
1.0
(b) Crack Path Deviation Profile
Trajectory Deviation Metrics :
• Initiation ($y \in [0.4, 0.5]$ mm) : $|\Delta x| \leq 3.5\,\mu$m ($l_0/4.3$)
• Mean Absolute Deviation (MAD) : $9.04\,\mu$m ($0.60\,l_0$)
• Root Mean Square (RMS) : $11.25\,\mu$m ($0.75\,l_0$)
• Max Path Deviation : $21.35\,\mu$m ($1.42\,l_0$)
• Trajectory Angle : $\theta = -59.35^\circ$ ($R^2 = 0.998$)
Horizontal Deviation $\Delta x = x_{\text{sim}} - x_{\text{lit}}$
Corridor Semi-Width ($\pm 120\,\mu$m)
Vertical Coordinate y (mm)
Lateral Deviation Δx (μm)
0.0
0.1
0.2
0.3
0.4
0.5
−150
−100
−50
0
50
100
150